Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 4: Modular Athematic

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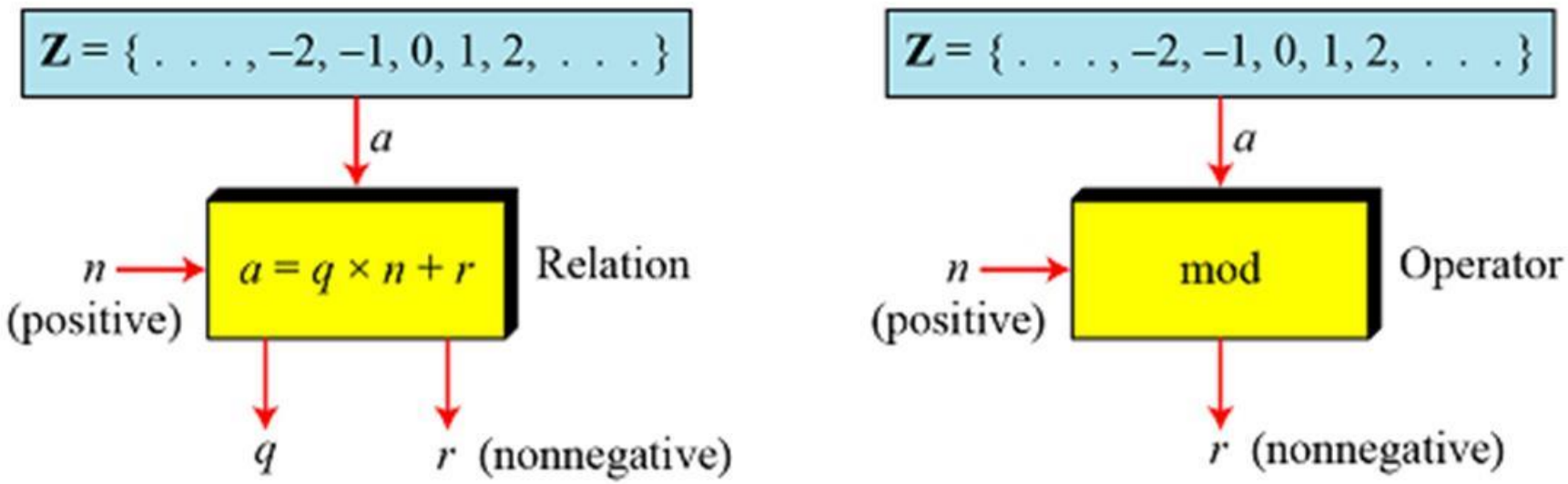
Preliminary

- The division relationship ($a = q \times n + r$) discussed in the previous section has two inputs (a and n) and two outputs (q and r).
- In modular arithmetic, we are interested in only one of the outputs, the remainder r.
- We use modular arithmetic in our daily life;
  - for example, we use a clock to measure time.
  - Our clock system uses modulo 12 arithmetic. However, instead of a 0 we use the number 12



Modulo Operator

- The modulo operator is shown as mod.
- The second input (n) is called the modulus.
- The output r is called the residue.



Division algorithm and modulo operator

Modulo Operator(cont.)

- Find the result of the following operations:
  a. 27 mod 5
  b. 36 mod 12
  c. −18 mod 14
  d. −7 mod 10
- Solution
  a. Dividing 27 by 5 results in r = 2
  b. Dividing 36 by 12 results in r = 0
  c. Dividing −18 by 14 results in r = −4. After adding the modulus r = 10
  d. Dividing −7 by 10 results in r = −7. After adding the modulus to −7, r = 3

Set of Residues

- The modulo operation creates a set, which in modular arithmetic is referred to as the set of least residues modulo n, or Z_n.

$$\mathbf{Z}_n = \{ 0, 1, 2, 3, \ldots, (n-1) \}$$

$$\mathbf{Z}_2 = \{ 0, 1 \} \qquad \mathbf{Z}_6 = \{ 0, 1, 2, 3, 4, 5 \} \qquad \mathbf{Z}_{11} = \{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \}$$

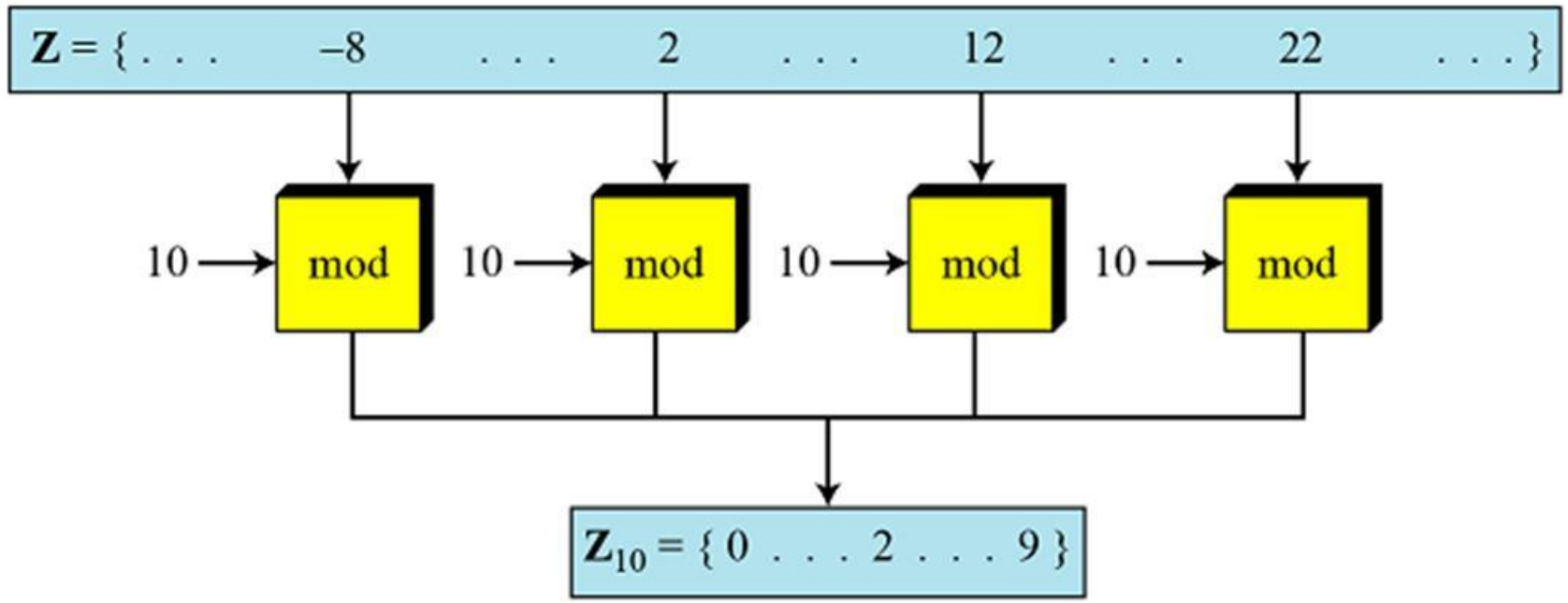
Some Z_n sets

Congruence

- To show that two integers are congruent, we use the congruence operator (≡).
- For example, we write:

$$2 \equiv 12 \pmod{10} \qquad 13 \equiv 23 \pmod{10}$$
$$3 \equiv 8 \pmod{5} \qquad 8 \equiv 13 \pmod{5}$$

Congruence



$$-8 \equiv 2 \equiv 12 \equiv 22 \pmod{10}$$

Congruence Relationship

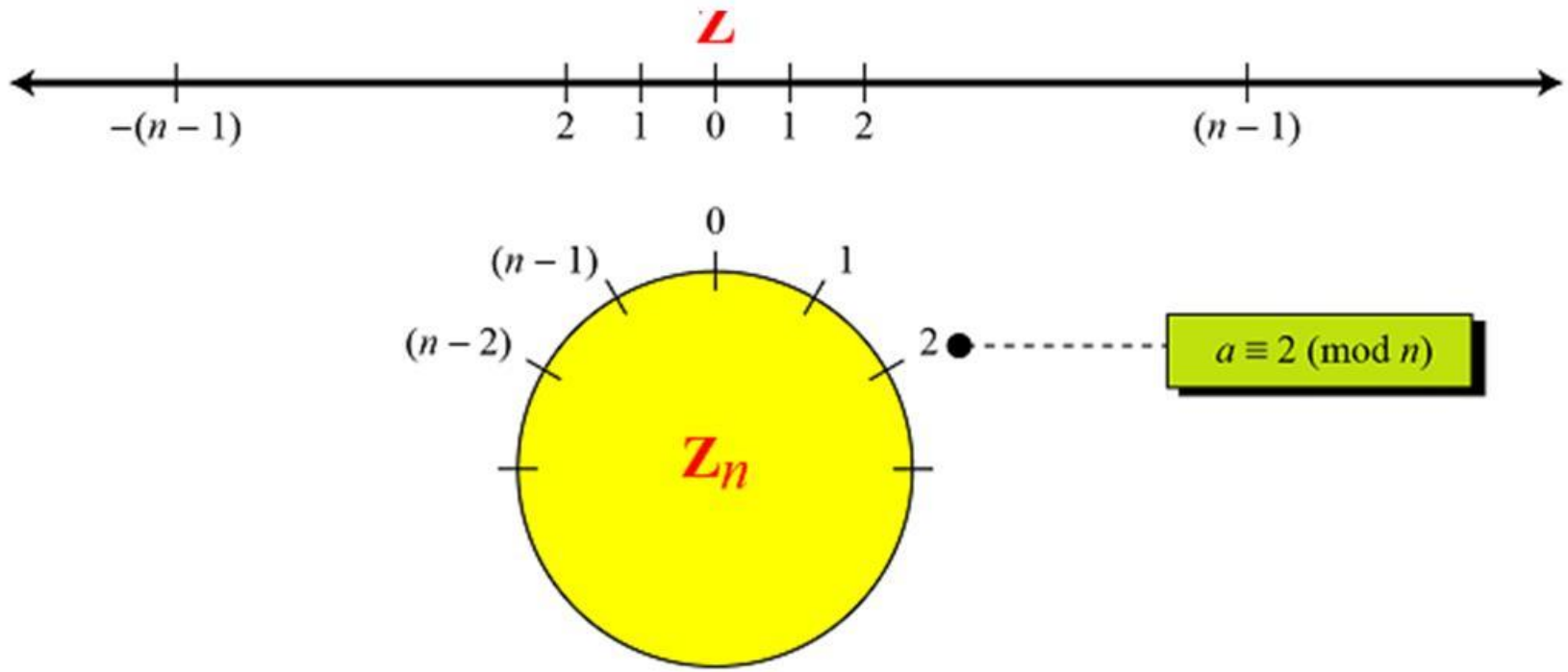
Congruence

- Residue Classes
  - A residue class [a] or $[a]_n$ is the set of integers congruent modulo n.
  - It is the set of all integers such that x=a(mod)n
  - E.g. for n=5, we have five sets as shown below:

$$
\begin{aligned}
[0] &= \{\ldots, -15, -10, -5, 0, 5, 10, 15, \ldots\} \\
[1] &= \{\ldots, -14, -9, -4, 1, 6, 11, 16, \ldots\} \\
[2] &= \{\ldots, -13, -8, -3, 2, 7, 12, 17, \ldots\} \\
[3] &= \{\ldots, -12, -7, -5, 3, 8, 13, 18, \ldots\} \\
[4] &= \{\ldots, -11, -6, -1, 4, 9, 14, 19, \ldots\}
\end{aligned}
$$

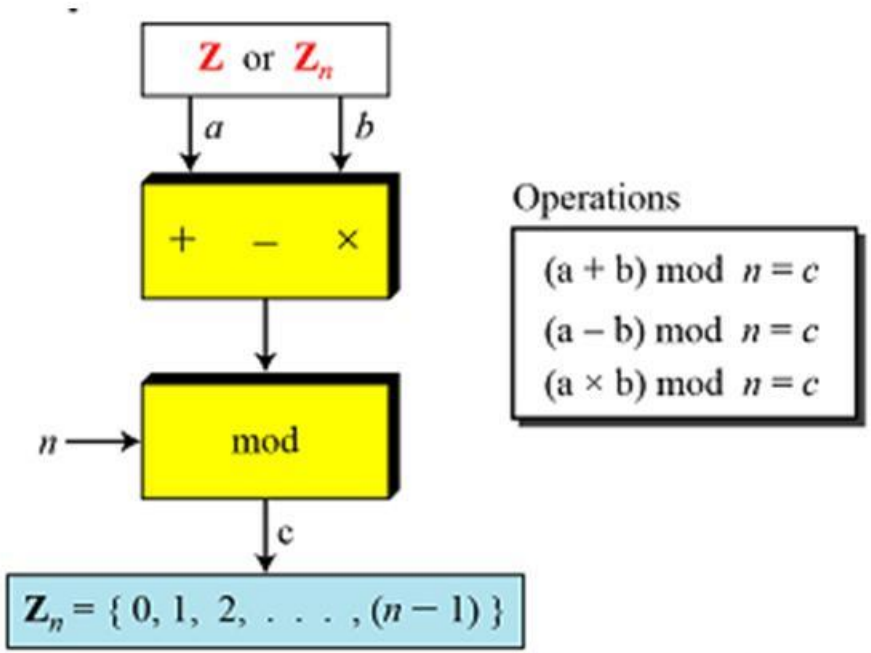
Congruence

- Comparison of Z and Z_n using graphs



Operation in Z_n

- The three binary operations that we discussed for the set Z can also be defined for the set Z_n.
- The result may need to be mapped to Z_n using the mod operator.



Operation in Z_n

- Perform the following operations (the inputs come from Z_n):
  a. Add 7 to 14 in Z_{15}.
  b. Subtract 11 from 7 in Z_{13}.
  c. Multiply 11 by 7 in Z_{20}.
- Solution

$$(14 + 7) \bmod 15 \rightarrow (21) \bmod 15 = 6$$
$$(7 - 11) \bmod 13 \rightarrow (-4) \bmod 13 = 9$$
$$(7 \times 11) \bmod 20 \rightarrow (77) \bmod 20 = 17$$

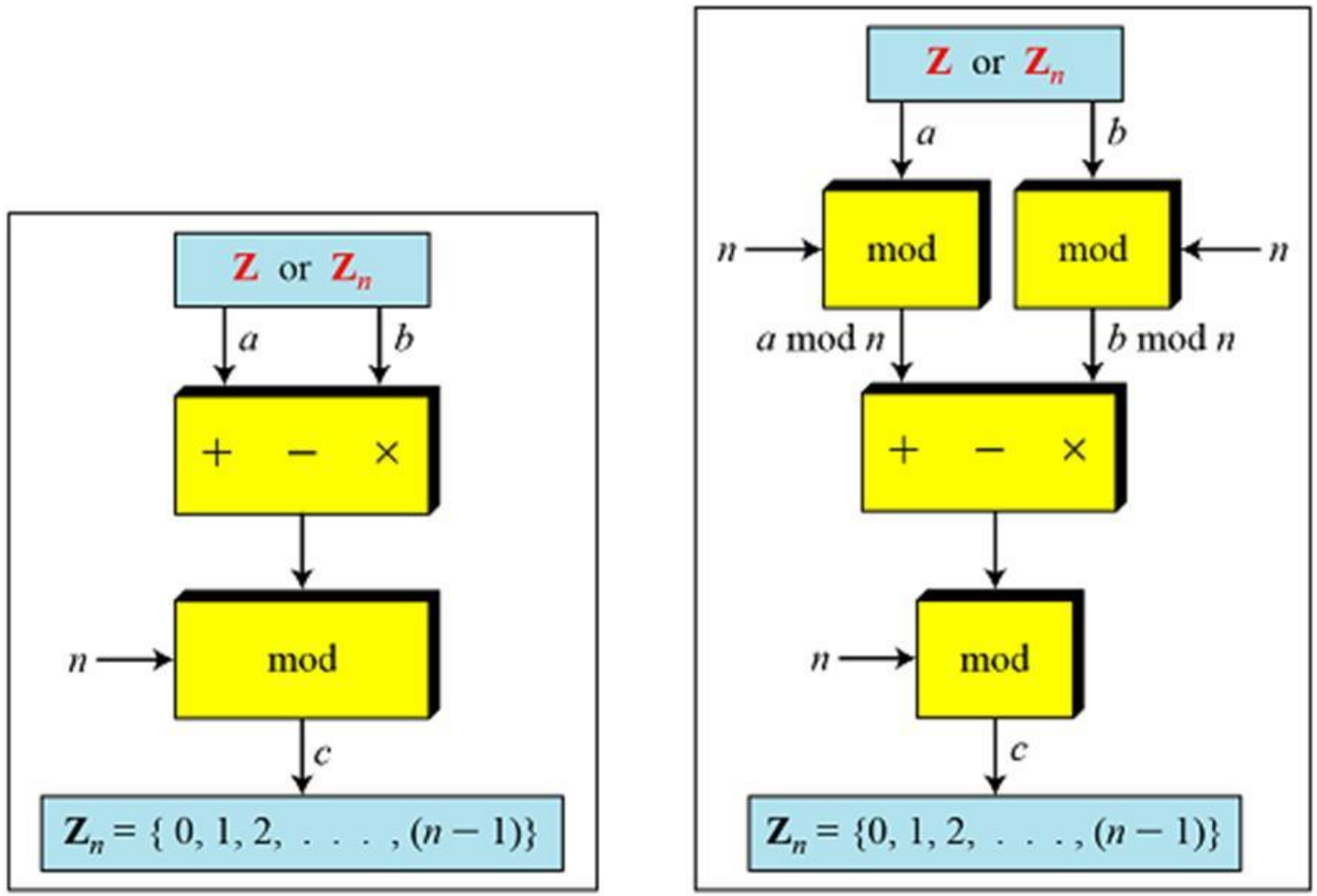
Operation in Z_n

- Perform the following operations (the inputs come from either Z or Z_n):
  a. Add 17 to 27 in Z_{14}.
  b. Subtract 43 from 12 in Z_{13}.
  c. Multiply 123 by −10 in Z_{19}.
- Solution
  a. Add 17 to 27 in Z_{14}: (17+27)mod 14 = 2
  - Subtract 43 from 12 in Z_{13} : (12-43)mod 13 = 8
  - Multiply 123 by −10 in Z_{19}. : (123 x (-10)) mod 19 = 5

Operation in Z_n(cont.)

First Property:	$(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$
Second Property:	$(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$
Third Property:	$(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

Operation in Z_n(cont.)



a. Original process b. Applying properties

Operation in Z_n(cont.)

- The following shows the application of the above properties:

  1. $(1{,}723{,}345 + 2{,}124{,}945) \bmod 11 = (8 + 9) \bmod 11 = 6$
  2. $(1{,}723{,}345 - 2{,}124{,}945) \bmod 11 = (8 - 9) \bmod 11 = 10$
  3. $(1{,}723{,}345 \times 2{,}124{,}945) \bmod 11 = (8 \times 9) \bmod 11 = 6$

Operation in Z_n(cont.)

- In arithmetic, we often need to find the remainder of powers of 10 when divided by an integer.

$$10^n \bmod x = (10 \bmod x)^n \qquad \text{Applying the third property } n \text{ times.}$$

$$\begin{aligned}
10 \bmod 3 = 1 \quad &\rightarrow \quad 10^n \bmod 3 = (10 \bmod 3)^n = 1 \\
10 \bmod 9 = 1 \quad &\rightarrow \quad 10^n \bmod 9 = (10 \bmod 9)^n = 1 \\
10 \bmod 7 = 3 \quad &\rightarrow \quad 10^n \bmod 7 = (10 \bmod 7)^n = 3^n \bmod 7
\end{aligned}$$

Inverses

- When we are working in modular arithmetic, we often need to find the inverse of a number relative to an operation.
- We are normally looking for an additive inverse (relative to an addition operation) or a multiplicative inverse (relative to a multiplication operation).

Additive Inverses

- In Z_n, two numbers a and b are additive inverses of each other if

$$a + b \equiv 0 \pmod{n}$$

In modular arithmetic, each integer has an additive inverse. The sum of an integer and its additive inverse is congruent to 0 modulo n.

Additive Inverses

- Find all additive inverse pairs in Z_{10}.
- Solution
  – The six pairs of additive inverses are (0, 0), (1, 9), (2, 8), (3, 7), (4, 6), and (5, 5).

Multiplicative Inverses

- In Z_n, two numbers a and b are the multiplicative inverse of each other if,

$$a \times b \equiv 1 \pmod{n}$$

In modular arithmetic, an integer may or may not have a multiplicative inverse.
When it does, the product of the integer and its multiplicative inverse is congruent to 1 modulo n.

Multiplicative Inverses

- Find the multiplicative inverse of 8 in Z_{10}.
  - There is no multiplicative inverse because gcd (10, 8) = 2 ≠ 1.
  - In other words, we cannot find any number between 0 and 9 such that when
    multiplied by 8, the result is congruent to 1.
- Find all multiplicative inverses in Z_{10}.
  - There are only three pairs: (1, 1), (3, 7) and (9, 9).
  - The numbers 0, 2, 4, 5, 6, and 8 do not have a multiplicative inverse.

Multiplicative Inverses

- Find all multiplicative inverse pairs in Z_{11}.
- Solution
  - We have seven pairs: (1, 1), (2, 6), (3, 4), (5, 9), (7, 8), and (10, 10).

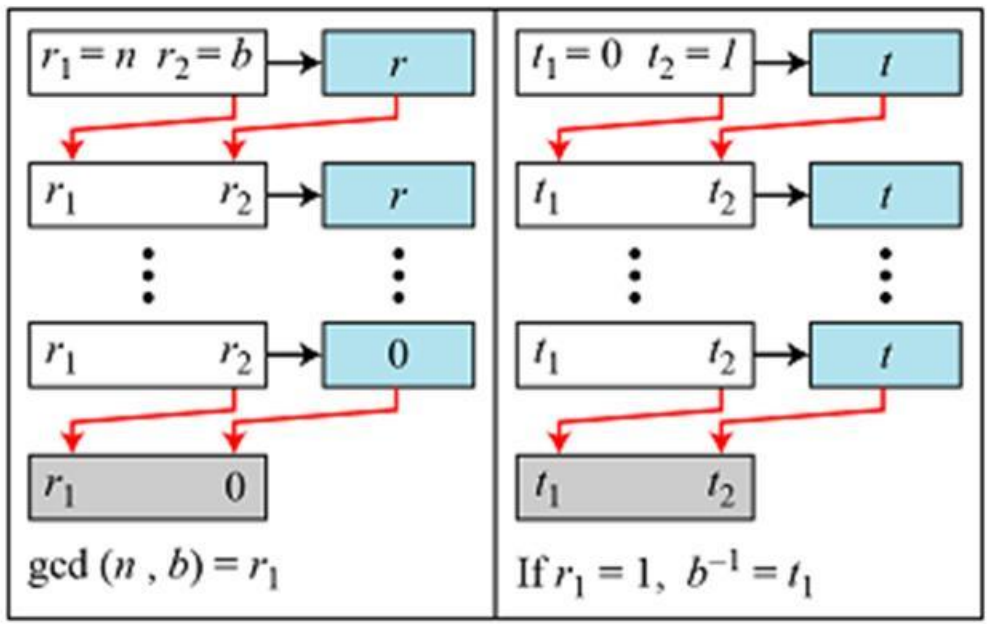
Multiplicative Inverses

- The extended Euclidean algorithm finds the multiplicative inverses of b in Z_n when n and b are given and gcd (n, b) = 1.
- The multiplicative inverse of b is the value of t after being mapped to Z_n.
- If the multiplicative inverse of b exists, gcd (n, b) must be 1.

$$(s \times n) + (b \times t) = 1$$

$$(s \times n + b \times t) \bmod n = 1 \bmod n$$
$$[(s \times n) \bmod n] + [(b \times t) \bmod n] = 1 \bmod n$$
$$0 + [(b \times t) \bmod n] = 1$$
$$(b \times t) \bmod n = 1 \quad \rightarrow \text{This means } t \text{ is the multiplicative inverse of } b \text{ in } Z_n$$

Multiplicative Inverses



a. Process

```
r1 ← n;      r2 ← b;
t1 ← 0;      t2 ← 1;
while (r2 > 0)
 {
  q ← r1 / r2;
  r  ← r1 – q × r2 ;
  r1 ← r2 ;      r2 ← r ;
  t  ← t1 – q × t2 ;
  t1 ← t2 ;      t2 ← t ;
 }
  if (r1 = 1) then b–1 ← t1
```

b. Algorithm

Using extended Euclidean algorithm to find multiplicative inverse

Multiplicative Inverses

- **Find the multiplicative inverse of 11 in Z_{26}.**

Solution

q	r_1	r_2	r	t_1	t_2	t
2	26	11	4	0	1	−2
2	11	4	3	1	−2	5
1	4	3	1	−2	5	−7
3	3	1	0	5	−7	26
	1	0		−7	26	

The gcd (26, 11) is 1; the inverse of 11 is -7 or 19.

Multiplicative Inverses

- Find the multiplicative inverse of 23 in Z_{100}.

Solution

q	r_1	r_2	r	t_1	t_2	t
4	100	23	8	0	1	−4
2	23	8	7	1	−4	19
1	8	7	1	−4	9	−13
7	7	1	0	9	−13	100
	1	0		−13	100	

The gcd (100, 23) is 1; the inverse of 23 is -13 or 87.

Multiplicative Inverses

- Find the inverse of 12 in Z_{26}.

Solution

q	r_1	r_2	r	t_1	t_2	t
2	26	12	2	0	1	–2
6	12	2	0	1	–2	13
	2	0		–2	13	

The gcd (26, 12) is 2; the inverse does not exist.

Addition and Multiplication Tables

- Addition and multiplication table for Z_{10}

+	0	1	2	3	4	5	6	7	8	9
0	**0**	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	**0**
2	2	3	4	5	6	7	8	9	**0**	1
3	3	4	5	6	7	8	9	**0**	1	2
4	4	5	6	7	8	9	**0**	1	2	3
5	5	6	7	8	9	**0**	1	2	3	4
6	6	7	8	9	**0**	1	2	3	4	5
7	7	8	9	**0**	1	2	3	4	5	6
8	8	9	**0**	1	2	3	4	5	6	7
9	9	**0**	1	2	3	4	5	6	7	8

Addition Table in $\mathbf{Z}_{10}$

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	**1**	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	**1**	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	**1**	8	0	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	**1**

Multiplication Table in $\mathbf{Z}_{10}$

Different Sets

- Some Z_n and Z_n* sets

$\mathbf{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

$\mathbf{Z}_6^* = \{1, 5\}$

$\mathbf{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$

$\mathbf{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$

$\mathbf{Z}_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$\mathbf{Z}_{10}^* = \{1, 3, 7, 9\}$

We need to use Zn when additive inverses are needed; we need to use Zn* when multiplicative inverses are needed.

Two More Sets

- Cryptography often uses two more sets: Z_p and Z_p*.
- The modulus in these two sets is a primenumber.

$$Z_{13} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$
$$Z_{13}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Linear Diophantine Equations

- **A linear Diophantine equation of two variables is,**
  $$ax + by = c.$$
- **We want to find integer values for x and y that satisfy the equation.**
- **Either no solution or an infinite number of solutions**
- **Let d = gcd(a,b); if $d \nmid c$, the equation has no solution.**
- **If $d \mid c$, the equation has infinite number of solutions : one of them is particular and the rest are general**

Linear Diophantine Equations(cont.)

Particular solution:

$x_0 = (c/d)s$ and $y_0 = (c/d)t$

General solutions:

$x = x_0 + k\,(b/d)$ and $y = y_0 - k(a/d)$

where k is an integer

Linear Diophantine Equations(cont.)

Example:

Find the particular and general solutions for the equation

$21x + 14y = 35.$

Linear Diophantine Equations(cont.)

- d = gcd (21, 14) = 7.
- Since 7| 35, the equation has an infinite number of solutions.
- We can divide both sides by 7 to find the equation 3x + 2y = 5.
- Using the extended Euclidean algorithm, we find s and t such as 3s + 2t = 1. We have s = 1 and t = −1.

Particular: $x_0 = 5 \times 1 = 5$ and $y_0 = 5 \times (-1) = -5$ since 35/7 = 5
General: $x = 5 + k \times 2$ and $y = -5 - k \times 3$ where k is an integer

Therefore, the solutions are (5, −5), (7, −8), (9, −11), . . . We can easily test that each of these solutions satisfies the original equation.

Linear Diophantine Equations(cont.)

Example:

Imagine we want to cash a Rs.100 cheque and get some Rs.20 notes and some Rs.5 notes.

Find out the possible choices if any exist for the given problem

Linear Diophantine Equations(cont.)

$d = \gcd(20, 5) = 5$ and $5 \mid 100$, infinite number of solutions.

$$4s + t = 1.$$

The particular solutions are $x_0 = 0 \times 20 = 0$ and $y_0 = 1 \times 20 = 20$.

$$(0, 20), (1, 16), (2, 12), (3, 8), (4, 4), (5, 0).$$